VirtualLoop

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Models

BIU Rack card mounted

Power: supplied by the cabinet backplane

Mounting: BIU rack plugin

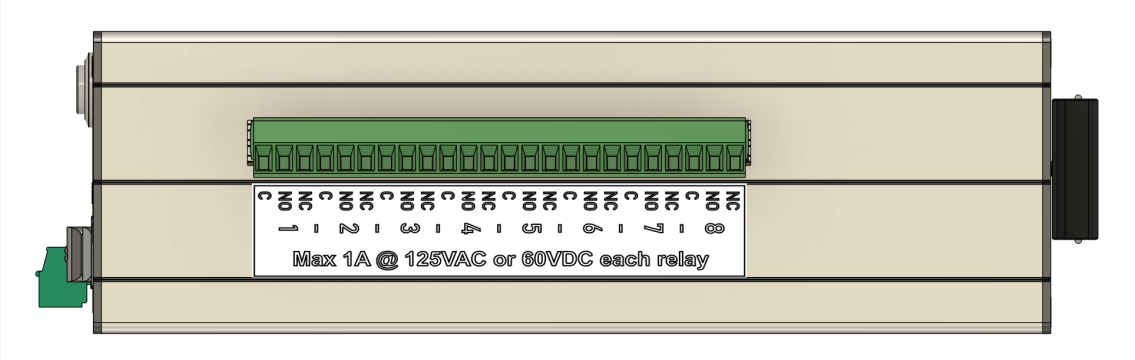
Relay Control: None

DIN rail mounted

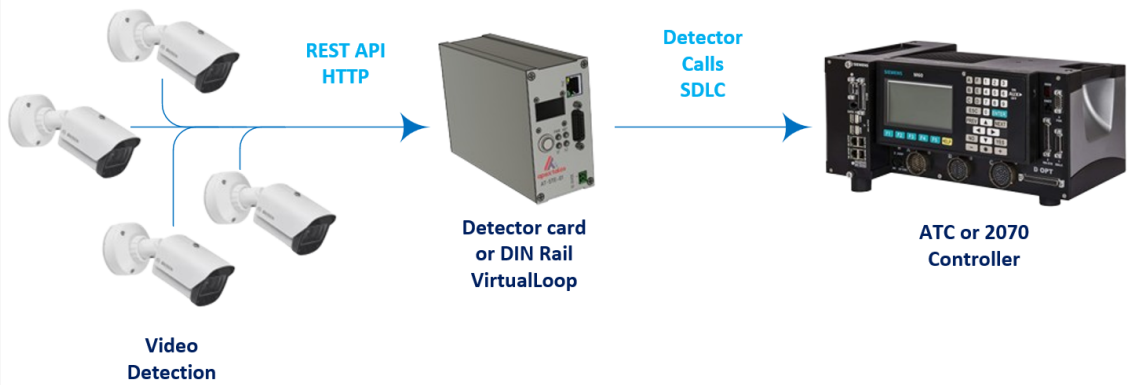
Power: supplied either by 12/24VDC on the front, or PoE 802.3af (consumes only ~1w)

Mounting: Din or shelf 2.1" wide x 6" deep x 4.1" high

Relay Control: 8 relays (C/NO/NC contacts) each can switch a max of 1A @ 125VAC or 60VDC Relays are numbered 1 through 8 with 1 closest to the front of the unit. Each relay includes Common, Normally Open, and Normally Closed contacts.



Product information and datasheets



Presentation on Intersection Detection and ATSPM Data Collection: [Intersection Video Detection.pptx]

VirtualLoop AT-STE-02 Datasheet: [VirtualLoop Datasheet.docx]

SDLC protocol reference: [NEMA TS2-2003.pdf]

Getting Started Checklist

- ☐ Installation
  - ☐ Power the device either through PoE, 12/24VDC plug, or BIU rack insertion
  - ☐ Verify the heartbeat light blinks 1 per second
- ☐ Networking
  - ☐ Setup a laptop to communicate on the subnet 192.168.1.XXX and plugin to the front Ethernet port
  - ☐ The device initial IP address is indicated by the OLED screen
  - ☐ Login to the unit's web interface using a browser
  - ☐ Set the proper address, subnet, and gateway IP for the network which allows communication to the cameras
- ☐ Detector integration
  - ☐ Plugin the SDLC interface cable
  - ☐ Use the VirtualLoop Web interface to enable the proper BIU #'s

    - ☐ Verify communication with the Intersection Control Unit through LEDs and exiting failsafe mode
    - ☐ Use the Bosch Camera Script Generator Tool to generate VCA tasks based on the rule and detector mappings
    - ☐ Install scripts onto the cameras as directed by the tool
- ☐ Additional Recommended Items
    - ☐ On the System tab, set a unique intersection name or number
    - ☐ Use the Firmware upgrades tool to install the latest firmware onto the device
    - ☐ Add failsafe monitoring configs
    - ☐ Add time synchronization configs
    - ☐ Rotate through the device screens checking for any errors

Factory Reset

Holding the front button for 5 seconds or longer will factory reset the device.

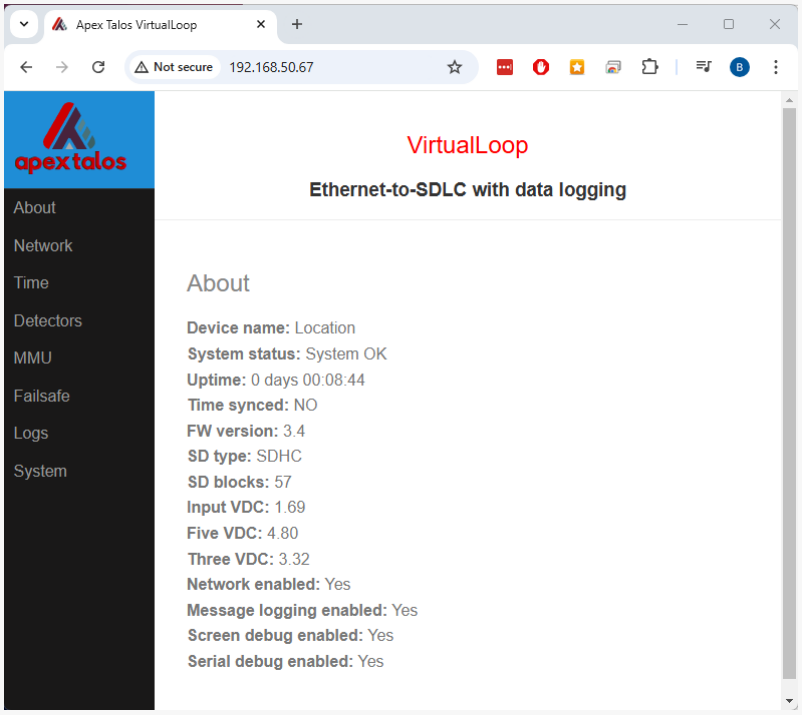
REST API

Supports placing calls on specific detector #, as well as other features

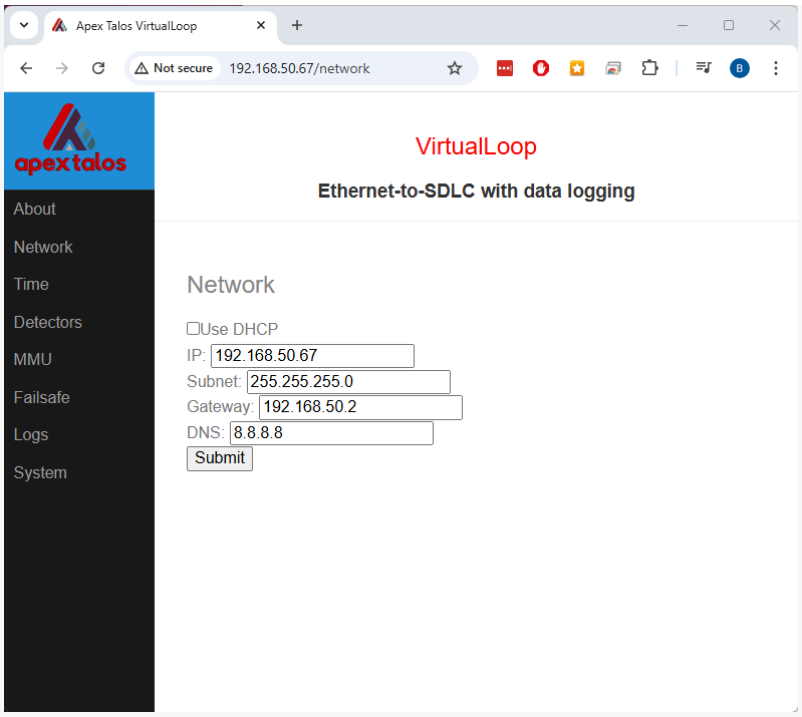
PostMan API tools: [PostMan]

Web Interface

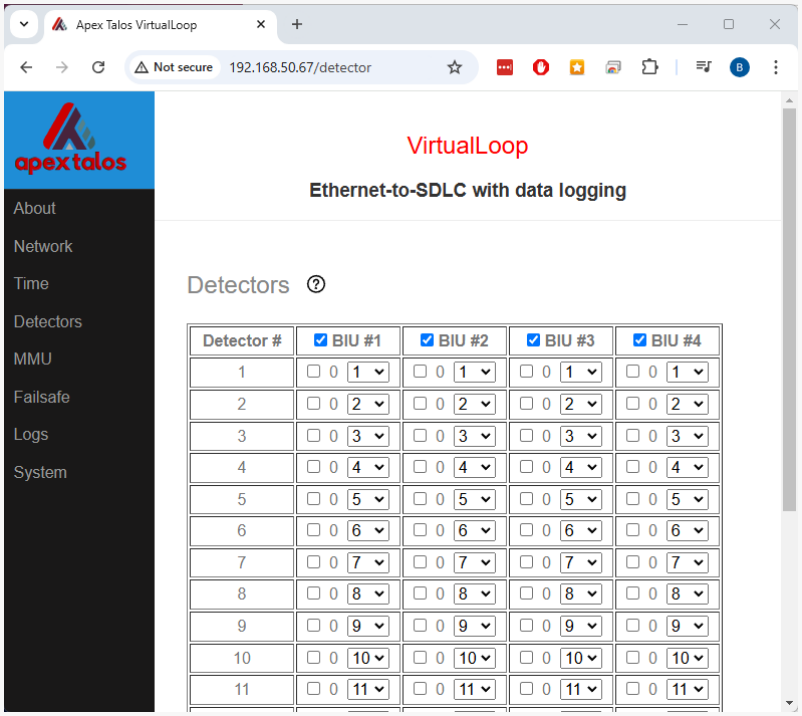
Status page - Reports version, internal voltage, device name, etc.



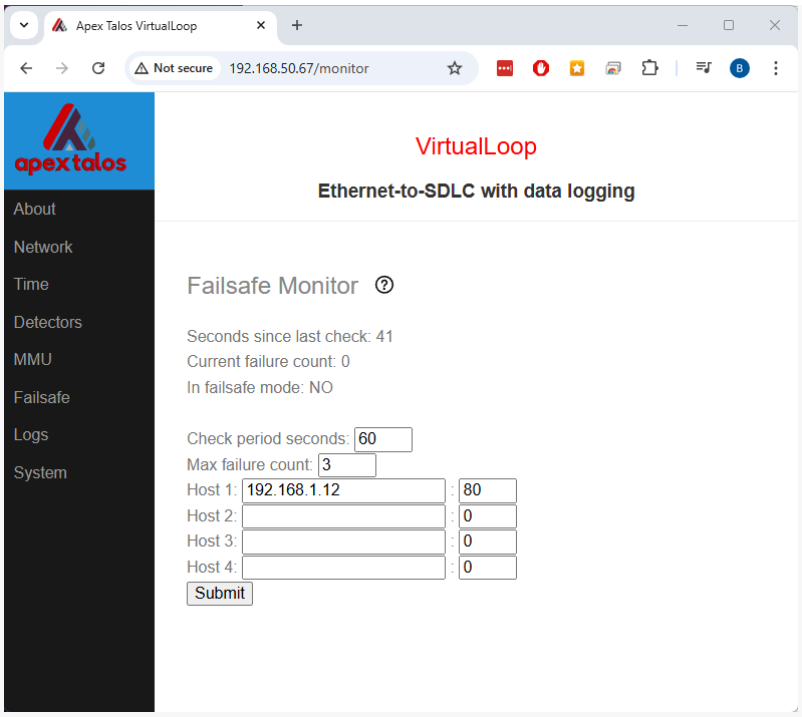
Network configuration page - Used to set the device IP through either DHCP or Static addressing



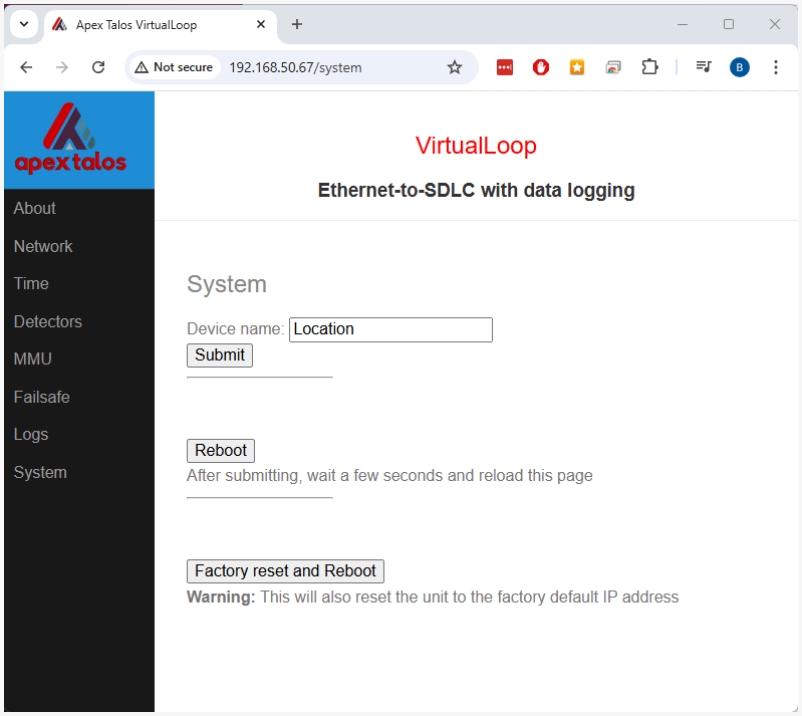
Detector selection page - Allows the user to force a detector on or off through the UI. Also reports the number of detector activation counts and provides the user a droplist of the load switch numbers for mapping in ATSPM data



Failsafe monitoring page - The device can monitor upto 4 IP address and ports. Typically these would be the camera detection devices. If the connection is not accepted within the specified number of seconds and failure counts, the VirtualLoop will go into FAILSAFE mode and automatically hold calls on all detectors.



System page - System functions like name, reboot, and factory reset.



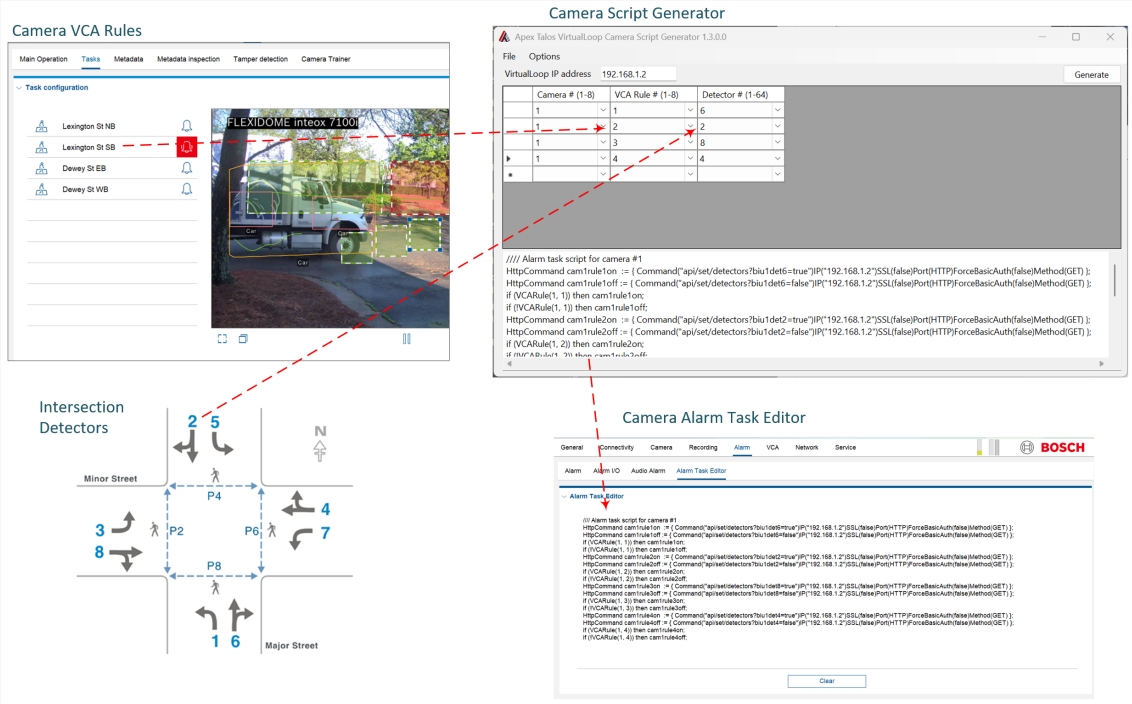
VirtualLoop Script Generator for Bosch Cameras

[VirtualLoop Camera Script Generator 1.6 Portable]

Generates the Bosch camera alarm task scripts for ease of use and to reduce errors.

1. Enter the IP address of the VirtualLoop device
2. Fill out the table with the mappings of cameras and rules to BIUs and detectors
3. Hit the "Generate" button and now you have each of the script blocks

4. Paste accordingly into your camera's Alarm Task Editors



Firmware Upgrades

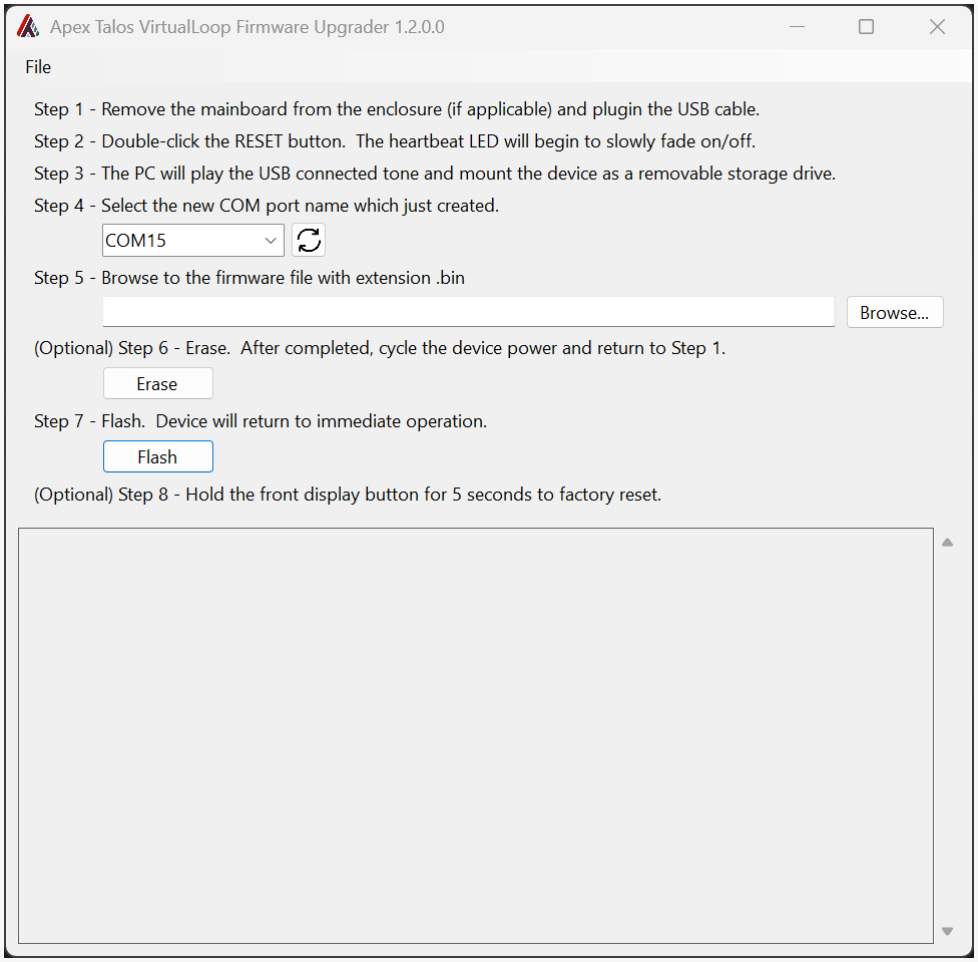
[VirtualLoop Firmware Upgrade Utility 1.2 Portable]

[VirtualLoop Firmware 4.0]

[VirtualLoop Firmware 3.9]

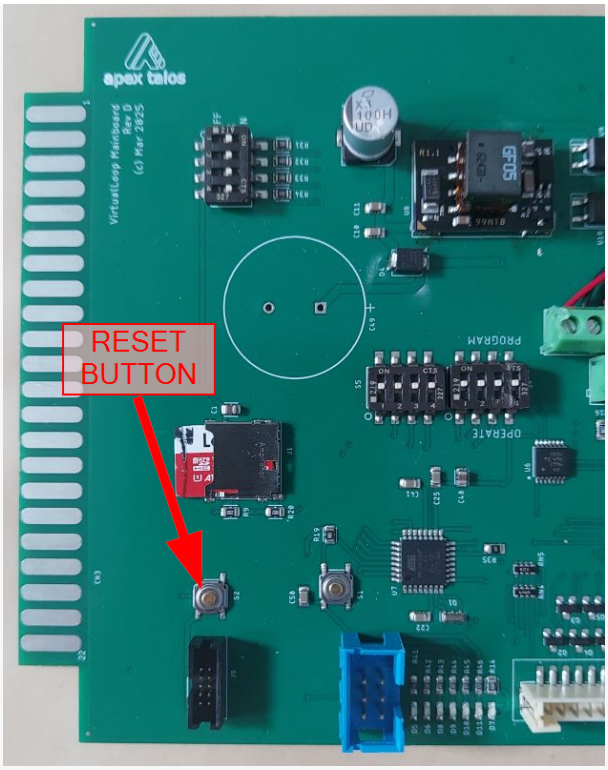
[VirtualLoop Firmware 3.8]

Download the firmware upgrade utility below and follow the instructions on the screen.



You'll need a USB micro cable like this:

Double-click the reset button:



Contact

For assistance, please contact Brandon at brandon@apextalos.tech